2026 Spring Semester
PHYS 263 - Data Visualization and analysis 6
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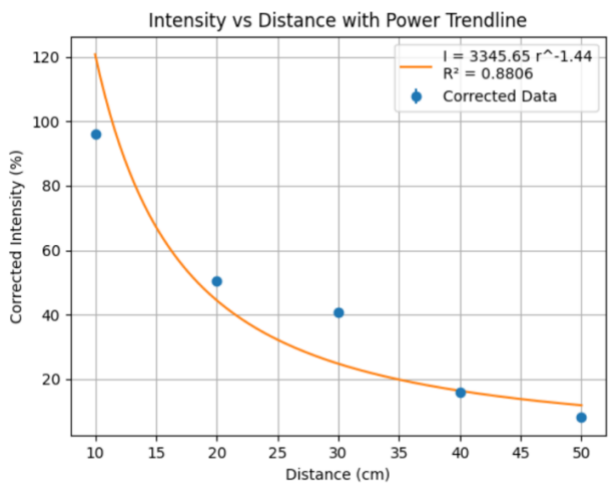
Data Visualization and Analysis

Table 1 - Measured and Corrected Light Intensity Data

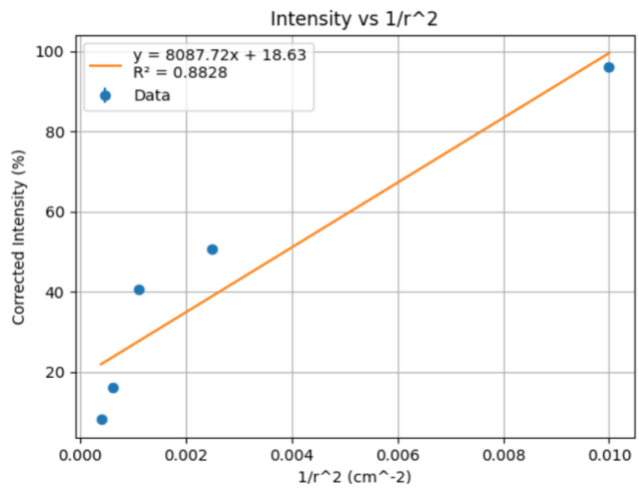
Distance (cm)	Measured (%)	Corrected (%)	$(1/r^2)$ (cm^{-2})
10	96.26	96.16	0.0100
20	50.66	50.56	0.0025
30	40.75	40.65	0.00111
40	16.10	16.00	0.000625
50	8.24	8.14	0.000400

Figure 1 - Intensity vs Distance



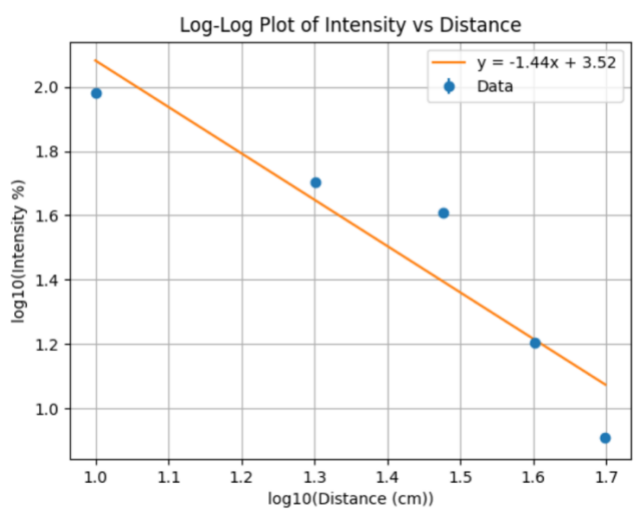
Observation: The light intensity decreases as distance increases in a nonlinear manner. The power trendline shows: $I = 3345.65 \cdot r\hat{}(-1.44)$. This indicates an inverse relationship between intensity and distance, consistent with theoretical expectations. The exponent (-1.44) is reasonably close to the expected value of -2, with deviation due to experimental error and non-ideal conditions such as background light and sensor alignment.

Figure 2 -Intensity vs 1/r²



Observation: The data forms an approximately straight line when plotted against $1/r$^2 , indicating a linear relationship. The trendline shows a positive linear relationship between intensity and $1/r$^2 , supporting the inverse square law. The fit is not perfect, which is reflected in the spread of the data points due to experimental uncertainties. This linearization demonstrates that the inverse square relationship is a good model for the experimental data. The trendline shows a moderate to strong fit, R ^2 $= 0.88$.

Figure 3 - Log-Log Plot



Observation: The log-log plot produces a nearly straight line. The slope is approximately -1.44 to -1.48, which is reasonably close to the expected value of -2. The deviation is due to experimental limitations such as background light, sensor alignment, and non-ideal point source behavior.

Trendline Analysis

A linear trendline was applied to the $1/r^2$ plot. The equation is: $E = 8087.72(1/r^2) + 18.63$. This relationship confirms that intensity is proportional to $1/r^2$, consistent with the inverse square law. The nonzero intercept is attributed to background light and sensor offset.

Uncertainty

The uncertainty in intensity is estimated as: $\Delta E = \pm 1\%$, based on light sensor variation. The uncertainty in distance is: $\Delta r = \pm 0.5"\ cm."$ This uncertainty affects the calculation of $1/r^2$, especially at smaller distances where small changes in r produce larger changes in $1/r^2$. Error bars representing intensity uncertainty are included on all graphs. These uncertainties contribute to the spread of data points and deviations from ideal linear behavior.

Final Observation

The results show that light intensity decreases with distance and follows an inverse square relationship. The linearized $1/r^2$ graph confirms the theoretical model, while small deviations are explained by measurement uncertainty and experimental limitations.